hw1

February 6, 2025

1 Homework 1

1.1 Question 1

s = 'hello 987' a) print(len(s)) returns 9

b) print(s[-3]) returns 9

c) print(s == "hello world") returns false

d) print(s.replace("hello", "world)) returns 'world 987'

e.1) '[]' are not used when calling a function which is what split is e.2) use 'split(" ")' instead

[47]:
```
# Import libraries
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
```

[48]:
```
# Question 1 Code

s = "hello 987"
print(len(s))
print(s[-3])
print(s == "hello world")
print(s.replace("hello", "world"))

# Incorrect
# words = s.split[" "]
# print(words)

# Correct
word = s.split(" ")
print(word)
```

```
9
9
False
world 987
['hello', '987']
```

1.2 Question 2

a) mean = 23.95 standard deviation = 16.735

b) median = 24 mode = 10

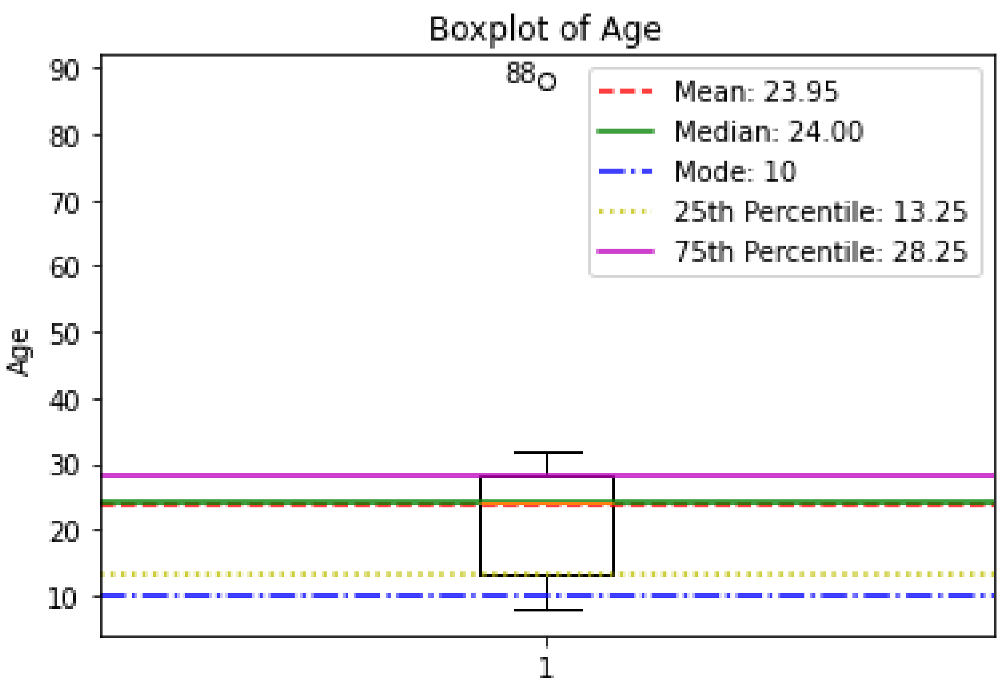
[52]:
```
# Question 2 Code

subjects = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19,
 ↪20]
age = [8, 10, 10, 10, 11, 14, 15, 15, 21, 23, 25, 25, 26, 27, 28, 29, 30, 32,
 ↪32, 88]

mean_age = np.mean(age)
std_age = np.std(age)
median_age = np.median(age)
mode_age = stats.mode(age)
percentile_25 = np.percentile(age, 25)
percentile_75 = np.percentile(age, 75)

print(f"Mean Age: {mean_age}")
print(f"Age Standard Deviation: {std_age}")
print(f"Median Age: {median_age}")
print(mode_age)
```

```
Mean Age: 23.95
Age Standard Deviation: 16.734619804465233
Median Age: 24.0
ModeResult(mode=10, count=3)
```

1.3 Question 2 Plots

[50]:
```
plt.boxplot(age)
plt.title("Boxplot of Age")
plt.ylabel("Age")

plt.axhline(mean_age, color="r", linestyle="--", label=f"Mean: {mean_age:.2f}")
plt.axhline(median_age, color="g", linestyle="-", label=f"Median: {median_age:.
 ↪2f}")
plt.axhline(mode_age.mode, color="b", linestyle="-.", label=f"Mode: {mode_age.
 ↪mode}")
plt.axhline(
    percentile_25,
    color="y",
    linestyle=":",
    label=f"25th Percentile: {percentile_25:.2f}",
)
plt.axhline(
    percentile_75,
```

```
    color="m",
    linestyle="-",
    label=f"75th Percentile: {percentile_75:.2f}",
)

# Label outliers
outliers = [y for y in age if y > mean_age + 2 * std_age or y < mean_age - 2 *␣
↪std_age]
for outlier in outliers:
    plt.text(0.99, outlier, f"{outlier}", horizontalalignment="right")

plt.legend()
plt.show()

# Identify outliers using the 3-sigma criterion
outliers_3sigma = [
    y for y in age if y > mean_age + 3 * std_age or y < mean_age - 3 * std_age
]
print("Outliers using 3-sigma criterion:", outliers_3sigma)
```




```
Outliers using 3-sigma criterion: [88]
```

1.4 Question 3

Nominal Objects | Color | Weather ——- | ——- | —– | 1 | Red | Sunny | 2 | Green | Cloudy| 3 | Red | Cloudy|

1.5 Steps to get proximity measure for the above dataset:

```
1. Assign binary attributes to each variable. For Color, Red is assigned 1 and Green assig

Nominal
```

Objects	Color(i)	Weather(j)
1	Red (1)	Sunny(1)
2	Green(0)	Cloudy(0)
3	Red (1)	Cloudy(0)

```
2. Calculate d(i, j) using the binary values.

- d(1,2) = 2 / 2 = 1
- d(1,3) = 1 / 2 = 0.5
- d(2,3) = 1 / 2 = 0.5
```

1.6 Question 4

Binary Objects | Education | Test | ——- | ———— | ——- | 1 | L | P | 2 | H | N | 3 | H | P |

Steps to get proximity measures for asymmetric binary variables

1. Assign 1 to H and 0 to L for Education. Assign 1 to P and 0 to N for Test

   Binary Objects | Education(i) | Test(j) ——- | ———— | ——- | 1 | L (0) | P (1)| 2 | H (1) | N (0)| 3 | H (1) | P (1)|

2. Calculate d(i, j)

```
d(1, 2) = 2 / 2 = 1
d(1, 3) = 1 / 2 = 0.5
d(2, 3) = 1 / 2 = 0.5
```

1.7 Question 5

```
| x | 0 | 2 | 0 | 0 | 0 | 6 | 0 | 3 | 0 | 0 |
| y | 3 | 1 | 0 | 0 | 1 | 0 | 0 | 4 | 0 | 2 |
```

Cosine Similarity

[51]:
```python
x = [0, 2, 0, 0, 0, 6, 0, 3, 0, 0]
y = [3, 1, 0, 0, 1, 0, 0, 4, 0, 2]
xy = np.dot(x, y)
print(f"Dot Product of x and y: {xy}")

vector_x = np.sqrt(sum(num**2 for num in x))
```

```
vector_y = np.sqrt(sum(num**2 for num in y))

cosine_similarity = xy / (vector_x * vector_y)
print("Cosine Similarity of x and y: ")
print(f"{xy} / ({vector_x} * {vector_y}) = {cosine_similarity}")
```

```
Dot Product of x and y: 14
Cosine Similarity of x and y:
14 / (7.0 * 5.5677643628300215) = 0.3592106040535498
```